

Assignment 3 – Jenkins Declarative Pipeline

Assignment 2 redeployed using a Jenkins Pipeline

1. Objective

Assignment 3 performs the same deployment work as Assignment 2, but instead of using three separate Jenkins Freestyle jobs, the complete process is defined in one Declarative Pipeline using a Jenkinsfile.

2. What Was Updated

Only the Jenkins execution method is changed. The Assignment 2 repository structure and deployment logic remain the same.

- Assignment 2: three separate Freestyle jobs.
- Assignment 3: one Jenkins Pipeline with sequential stages.
- The existing repo1, repo2 and repo3 content is reused.
- The new Jenkinsfile is added to the repository and executed by Jenkins.

3. Prerequisites

- Jenkins is installed and running.
- Jenkins has access to the GitHub repository.
- The EC2 instance has /var/www/html available.
- The repository contains jenkins_ec2_assig2/repo1, repo2 and repo3.
- The Jenkins agent has permission to create/copy files under /var/www/html.

4. Repository Structure

Expected structure:

```
Jenkins-assignment/  
├── jenkins_ec2_assig2/  
│   ├── repo1/  
│   │   └── index.html  
│   ├── repo2/  
│   │   └── index.html  
│   └── repo3/  
│       └── index.html  
└── Jenkinsfile
```

5. Jenkinsfile

```
pipeline {  
    agent any  
  
    stages {  
        stage('Checkout') {  
            steps {  
                git branch: 'main',  
                    url: 'https://github.com/DV-45/Jenkins-assignment.git'  
            }  
        }  
    }  
}
```

```

    }
}

stage('Q1 Release') {
    steps {
        sh '''
        mkdir -p /var/www/html/R1
        cp -r jenkins_ec2_assig2/repo1/* /var/www/html/R1/
        ls -la /var/www/html/R1
        '''
    }
}

stage('Q4.2 Patch') {
    steps {
        sh '''
        mkdir -p /var/www/html/R2
        cp -r jenkins_ec2_assig2/repo2/* /var/www/html/R2/
        ls -la /var/www/html/R2
        '''
    }
}

stage('Q3.3 Bug Fix') {
    steps {
        sh '''
        mkdir -p /var/www/html/R3
        cp -r jenkins_ec2_assig2/repo3/* /var/www/html/R3/
        ls -la /var/www/html/R3
        '''
    }
}
}
}
}
}

```

6. Add Jenkinsfile to GitHub

1. Open the local cloned Jenkins-assignment repository.
2. Create a file named Jenkinsfile in the repository root.
3. Paste the pipeline code from Section 5.
4. Save the file.
5. Run `git status` to verify the new file.
6. Run `git add Jenkinsfile`.
7. Commit the change.
8. Push the change to the main branch.

Suggested Git commands:

```

git status
git add Jenkinsfile
git commit -m "Add Assignment 3 Jenkins Pipeline"
git push origin main

```

Screenshot 1 – GitHub Repository with Jenkinsfile

Capture the GitHub repository showing Jenkinsfile and the existing jenkins_ec2_assig2 folder.

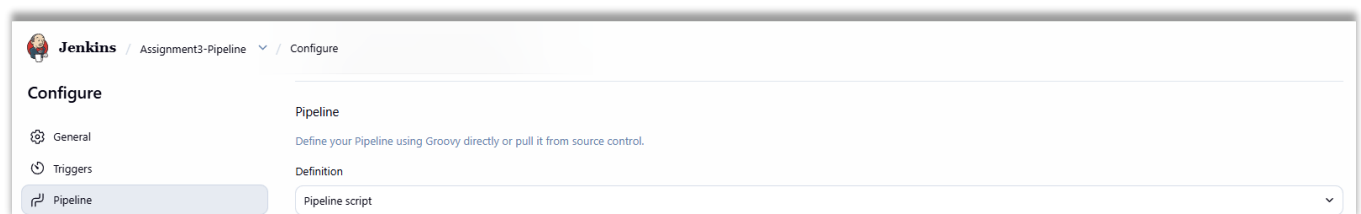
[SCREENSHOT SPACE – GitHub repository]

7. Create Jenkins Pipeline Job

1. Open Jenkins.
2. Click New Item.
3. Enter a job name, for example: Jenkins-Assignment-3-Pipeline.
4. Select Pipeline.
5. Click OK.
6. Scroll to the Pipeline section.
7. For Definition, select Pipeline script from SCM.
8. Select Git as SCM.
9. Enter the repository URL: <https://github.com/DV-45/Jenkins-assignment.git>
10. Select branch: */main.
11. Set Script Path to Jenkinsfile.
12. Click Save.

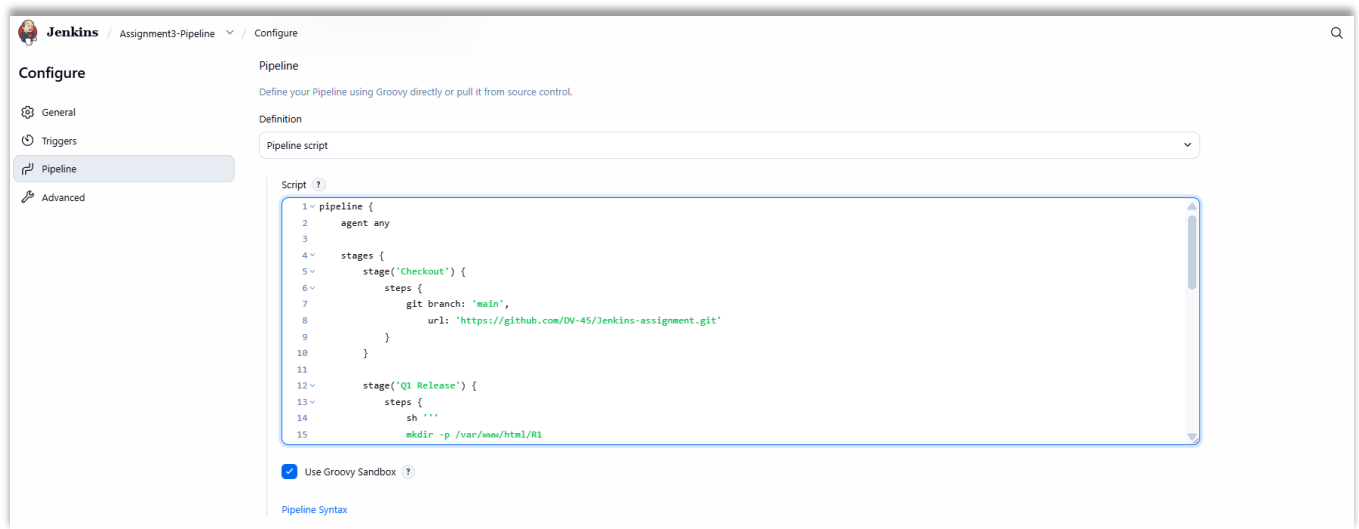
Screenshot 2 – Create Pipeline Job

Capture the New Item page showing the Pipeline job type.



Screenshot 3 – Pipeline SCM Configuration

Capture the Pipeline configuration showing Git repository, main branch and Jenkinsfile path.

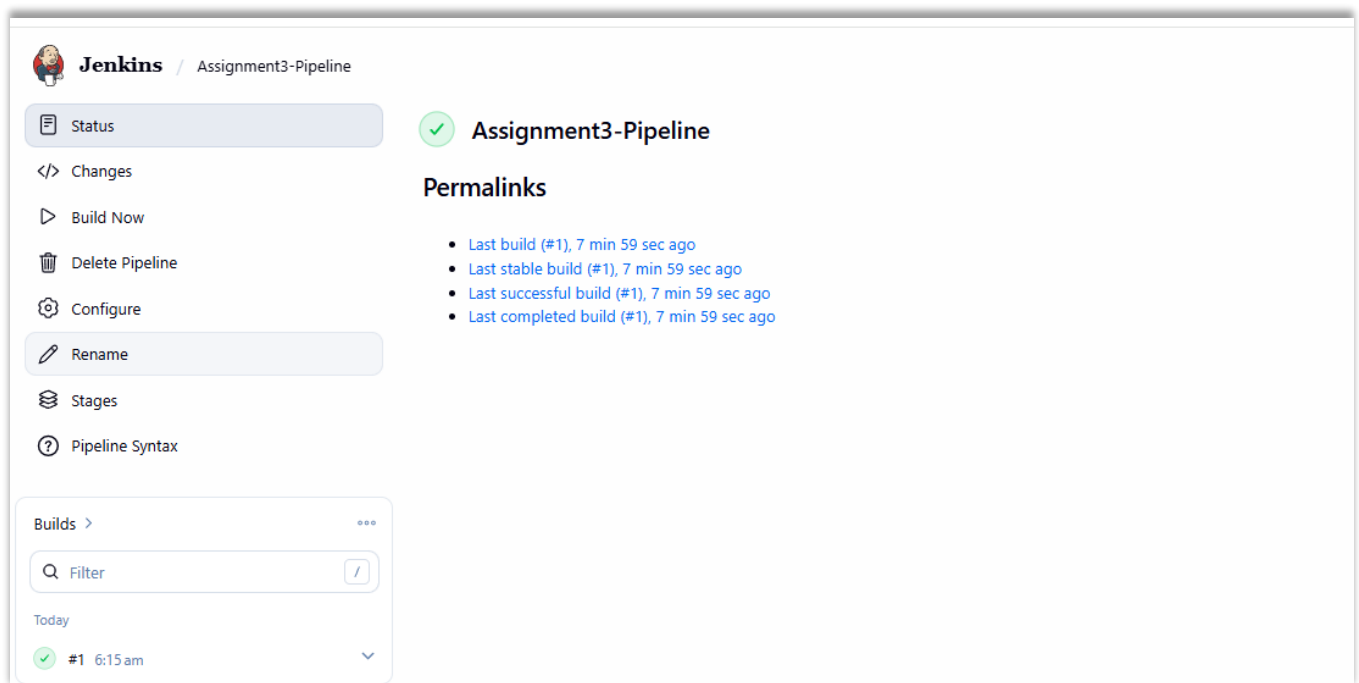


8. Run the Pipeline

1. Open the Jenkins-Assignment-3-Pipeline job.
2. Click Build Now.
3. Open the latest build number.
4. Click Console Output.
5. Observe the stages executing in order: Checkout → Q1 Release → Q4.2 Patch → Q3.3 Bug Fix.
6. Wait for Finished: SUCCESS.

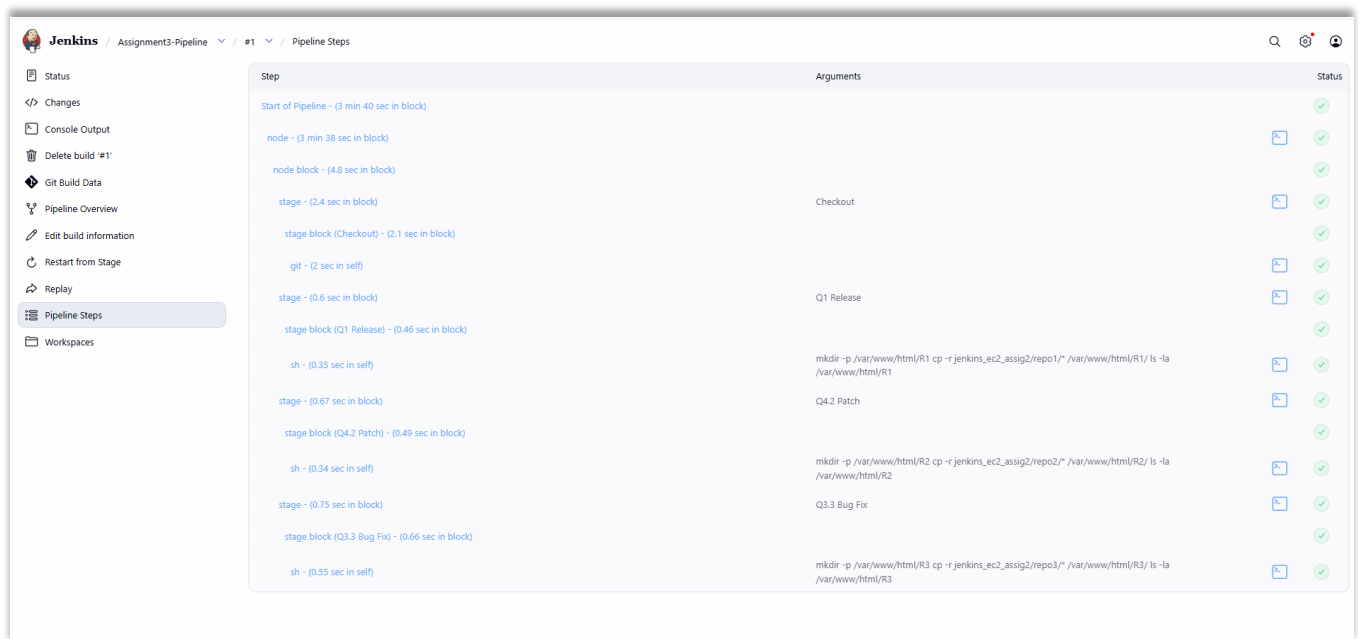
Screenshot 4 – Pipeline Build

Capture the Jenkins build page showing the latest successful build.



Screenshot 5 – Pipeline Stage View

Capture the stage visualization showing Checkout, Q1 Release, Q4.2 Patch and Q3.3 Bug Fix.



Screenshot 6 – Console Output

Capture Console Output showing the copy commands and directory listings.



Status

<> Changes

📄 Console Output

🗑️ Delete build '#1'

🔑 Git Build Data

🔗 Pipeline Overview

✎ Edit build information

🔄 Restart from Stage

↺ Replay

☰ Pipeline Steps

📁 Workspaces

Console

📄 Download

📄 Copy

View as plain text

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Still waiting to schedule task
Waiting for next available executor
Pausing (shutting down)
Resuming build at Sat Oct 03 06:19:16 UTC 2026 after Jenkins restart
Ready to run at Sat Oct 03 06:19:16 UTC 2026
Running on Jenkins in /var/lib/jenkins/workspace/Assignment3-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/DV-45/Jenkins-assignment.git
> git init /var/lib/jenkins/workspace/Assignment3-Pipeline # timeout=10
Fetching upstream changes from https://github.com/DV-45/Jenkins-assignment.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/DV-45/Jenkins-assignment.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/DV-45/Jenkins-assignment.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision f318a966770acd1da718d779958a9b2d6ca694b (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f f318a966770acd1da718d779958a9b2d6ca694b # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b main f318a966770acd1da718d779958a9b2d6ca694b # timeout=10
Commit message: "added guide and readme file"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Q1 Release)
[Pipeline] sh
+ mkdir -p /var/www/html/R1
+ cp -r jenkins_ec2_assig2/repo1/index.html /var/www/html/R1/
+ ls -la /var/www/html/R1
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 06:19 index.html
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Q4.2 Patch)
[Pipeline] sh
+ mkdir -p /var/www/html/R2
+ cp -r jenkins_ec2_assig2/repo2/index.html /var/www/html/R2/
+ ls -la /var/www/html/R2
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 06:19 index.html
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Q3.3 Bug Fix)
[Pipeline] sh
+ mkdir -p /var/www/html/R3
+ cp -r jenkins_ec2_assig2/repo3/index.html /var/www/html/R3/
+ ls -la /var/www/html/R3
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct 3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct 3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct 3 06:19 index.html
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

9. Verify Deployment on EC2

Run the following commands on the EC2 server:

```
ls -la /var/www/html/R1
ls -la /var/www/html/R2
ls -la /var/www/html/R3
```

Screenshot 7 – EC2 Deployment Directories

Capture the terminal showing R1, R2 and R3 and their deployed files.

```
[root@ip-172-31-1-51 R3]# ls -la /var/www/html/R1
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 06:19 index.html
[root@ip-172-31-1-51 R3]# ls -la /var/www/html/R2
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 06:19 index.html
[root@ip-172-31-1-51 R3]# ls -la /var/www/html/R3
total 4
drwxr-xr-x. 2 jenkins jenkins 24 Oct  3 06:19 .
drwxr-xr-x. 5 jenkins jenkins 36 Oct  3 03:19 ..
-rw-r--r--. 1 jenkins jenkins 56 Oct  3 06:19 index.html
[root@ip-172-31-1-51 R3]#
```

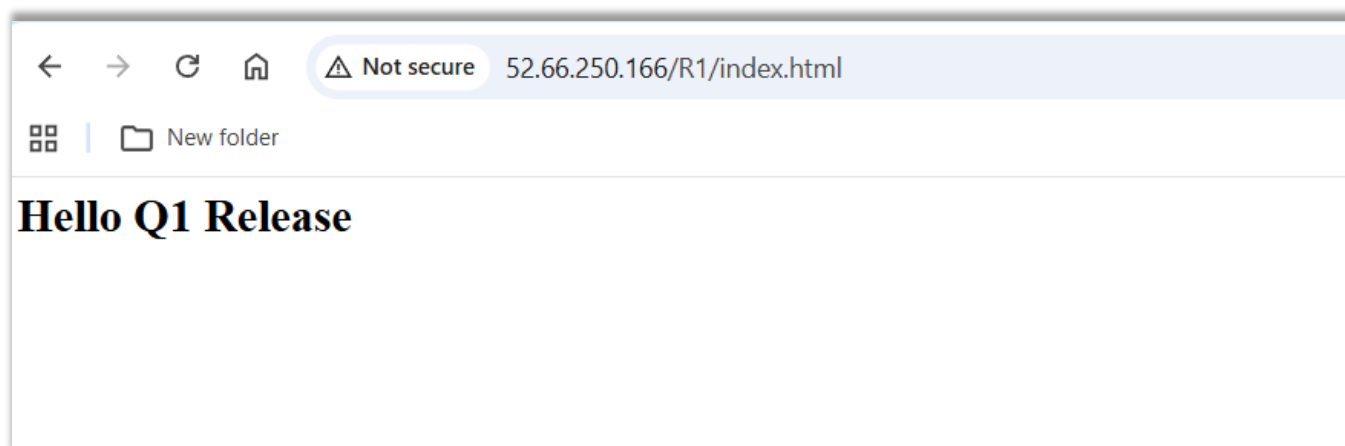
10. Verify in Browser

If Apache/httpd is serving /var/www/html, open:

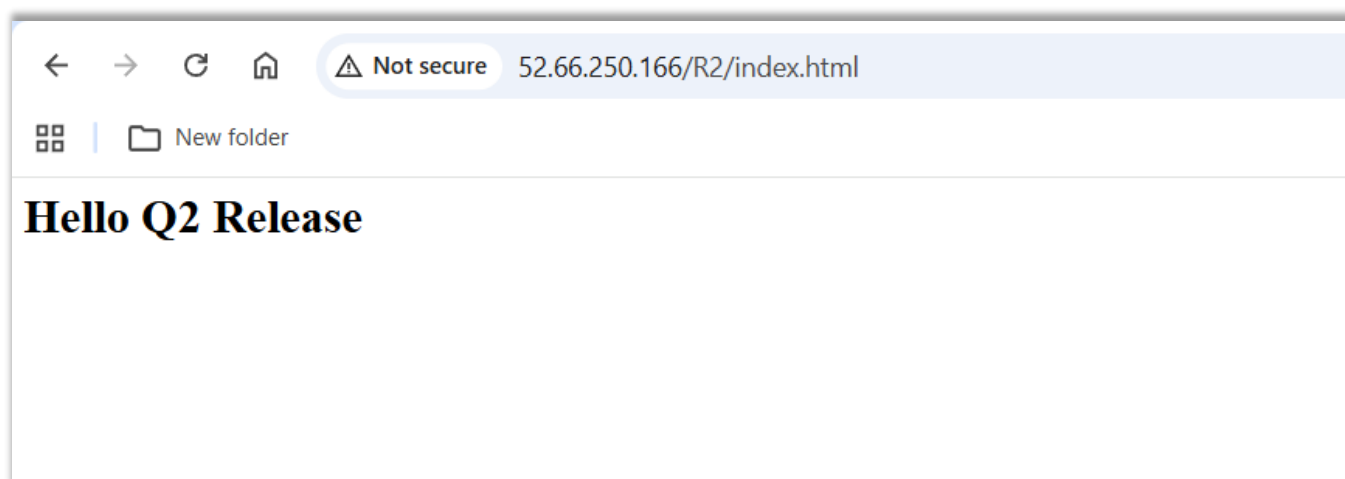
- <http://<EC2-PUBLIC-IP>/R1/>
- <http://<EC2-PUBLIC-IP>/R2/>
- <http://<EC2-PUBLIC-IP>/R3/>

Replace <EC2-PUBLIC-IP> with the public IP of your EC2 instance.

Screenshot 8 – R1 Browser Verification



Screenshot 9 – R2 Browser Verification



Screenshot 10 – R3 Browser Verification

